



Relax and Recover

Relax-and-Recover simplifies
Linux Disaster Recovery



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Who am I

- Independent Unix System Engineer since 1996
- Started with Unix in 1986; Linux since 1991
- Open Source contributor and maintainer:
 - Relax and Recover (rear)
 - Upgrade-UX
 - Config-to-HTML (cfg2html v6.x – HP-UX and Linux track)
 - Criux (node and cluster consistency checker)
 - More projects : <https://github.com/gdha>





Business Continuity





What is your Disaster Recovery Plan?



**Our Disaster Recovery Plan
Goes Something Like This...**





Disaster Recovery Concerns

- **Uptime**
 - Quick restores with minimal or no manual steps after the recovery
- **Trustworthy**
 - Avoid corrupted file systems and that system boots after recovery
- **Cost**
 - DR solutions need to be affordable
- **Complexity**
 - DR plans tend to be too complex.



Relax and Recover (rear) as DR solution

- Rear is a tool that implements a **DR work-flow** for Linux
- Basically meaning:
 - Modular framework written in Bash
 - Easy to extend to own needs
 - Easy to deploy (set up and forget)
 - Integration for various Linux technologies
 - Integration with various back-up solutions
 - Attempts to make system recovery *as easy as possible*
- Rear runs on-line (no downtime to create a DR image)



Relax and Recover – What is Rear?

- **<http://relax-and-recover.org/>**
- GPL Software – Developers in Germany and Belgium
- 100% Bash script – no GUI and no dependencies
- Utilize kernel, modules, binaries of host (kernel ≥ 2.6)
- Support any combination of SW/HW RAID, LVM
- Support many different file-systems (btrfs, ext*, xfs,...)
- Internal backup on CIFS, NFS, USB, rsync ...
- Boot media on CD/DVD, USB storage and LAN (PXE)
- Bootable tapes



Relax and Recover - Features

- Focus on disaster recovery and **not** backup
- Tight integration with common backup software – delegate file backup to backup infrastructure
- Simple *full* backup integrated (with tar or rsync)
- Complements backup software:
 - Backup software: Data storage and retrieval
 - Rear: Recover system layout and make it work again
 - Rear utilizes the backup software to restore the backup data
- Use the best tool for the job



Decide on DR strategy

- Which backup mechanism to use?
 - Internal backup: GNU tar, rsync
 - External backup: bacula, bareos, commercial backup solution
- Where will the backups reside?
 - NFS share, CIFS share, external USB disk, tape, local spare disk, cloud storage, DVD
 - Remote network and/or storage location
- How shall we boot the rescue image?
 - Via DVD (ISO image), tape (OBDR), network (PXE), USB disk



Disaster Recovery - Media

- Most important: External storage!
- Bootable media: CD/DVD, USB key, LAN, tape ...
- Media usually combination boot and backup media:
 - Bootable CD/DVD, USB key with backup data on it
 - LAN boot (PXE) with backup data via CIFS, NFS ...
 - Bootable tapes - HP OBDR (CD emulation)
- Separation between boot media and backup data
 - Boot the system from a (small) USB key, CD/DVD or LAN
 - Recover the system with backup software, tar, rsync ...



Disaster Recovery – How It Works

- Store the disk layout
 - Partitioning, LVM and RAID configuration
 - File systems, file system labels ...
 - Boot loader (GRUB, GRUB2, LILO, UEFI elilo/grub)
- Store the files (tar, rsync, through backup software ...)
- Create bootable rescue media with system configuration (and backup data)
- Can be done online
 - No business interruption
 - 100% compatible with original systems hard- and software



Disaster Recovery – Rescue Media

- Create “rescue linux” from running system
- Optimally compatible “tool box”
- Clone the system environment
 - Linux kernel and modules
 - Device driver configuration
 - Network configuration
 - Basic system software and tools
- Operate entirely in RAM (initrd)





Disaster Recovery – In Action

- Boot system from rescue media (into RAM)
- Restore disk layout
 - Create partitions, RAID configuration and LVM
 - Create file systems (mkfs, mkswap)
 - Configure file systems (labels, mount points)
- Restore the backup data
- Restore the boot loader
- Reboot
- Done within minutes



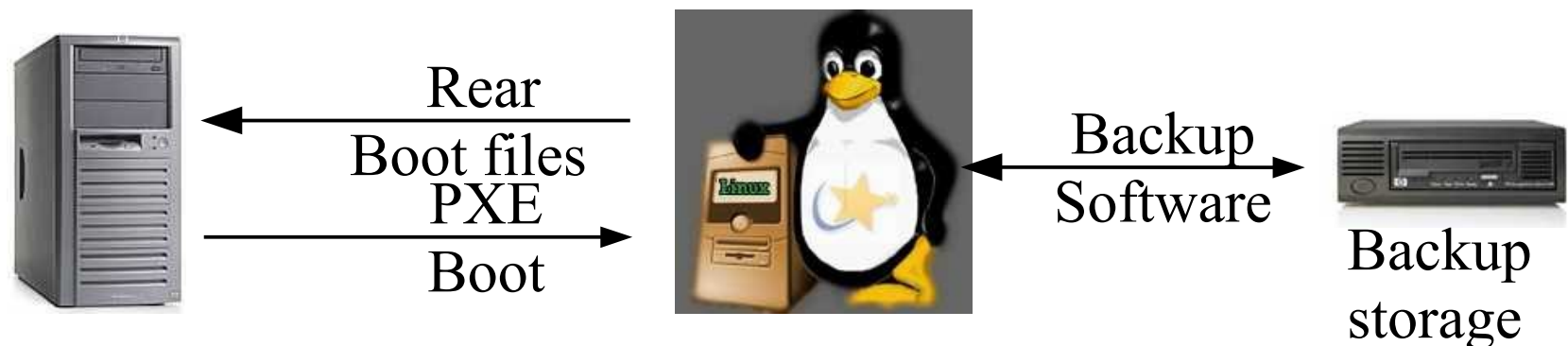
Relax and Recover – Backup Software

- Supported solutions include:
 - CommVault Galaxy; EMC² Networker (Legato)
 - IBM Tivoli Storage Manager
 - Symantec NetBackup; HP Data Protector
 - Bacula, Bareos
 - Duplicity
 - Rsync and other “external” methods
 - GNU tar archive on NAS share – CIFS, NFS, ...
- Very transparent integration
- Can be easily extended to support other backup vendors



Relax and Recover – Network Integration

- Disaster recovery as part of network infrastructure
 - Backup software – file-level backup storage
 - Rear – system environment
 - Boot rescue media via PXE – no physical media required
 - Very scalable – automated installation of entire disaster recovery data center
- Disaster Recovery Linux Manager (drlm.org) – kind of rear server





Getting started with Relax and Recover (rear)

- Download it from
 - Stable rear rpm's build from Github; via OpenSuse Build Service
 - <http://download.opensuse.org/repositories/Archiving:/Backup:/Rear/>
 - The rear-snapshot rpm's build from Github; via OBS
 - <http://download.opensuse.org/repositories/Archiving:/Backup:/Rear:/Snapshot/>
 - The official source
 - <https://github.com/rear/rear>
 - The official repo's (Fedora, EPEL, RHEL and SLES)
 - `yum install rear`
 - `zypper install rear`



Installation of rear

- E.g. on CentOS
`# yum install rear-1.17.2-111.git201512241143.el7.x86_64.rpm`
- We also need syslinux (if not present yet)
`# yum install syslinux`
- Install nfs-client, cifs-utils, rsync, syslinux-extlinux if required
- Do not forget openssh (should be there by default)



Relax and Recover - Backup Flows

- Internal **BACKUP=**
 - **NETFS**: NFS, CIFS, USB, TAPE, ISO, FILE
 - BACKUP_PROG=tar, or rsync (not for TAPE, ISO)
 - **RSYNC**: rsync method
 - BACKUP_PROG=rsync (to remote location)
- External **BACKUP=**
 - REQUESTRESTORE, EXTERNAL
 - BACULA, BAREOS (“open source” backup software)
 - DP, NBU, TSM, NSR, GALAXY (commercial backup)
 - DUPLICITY

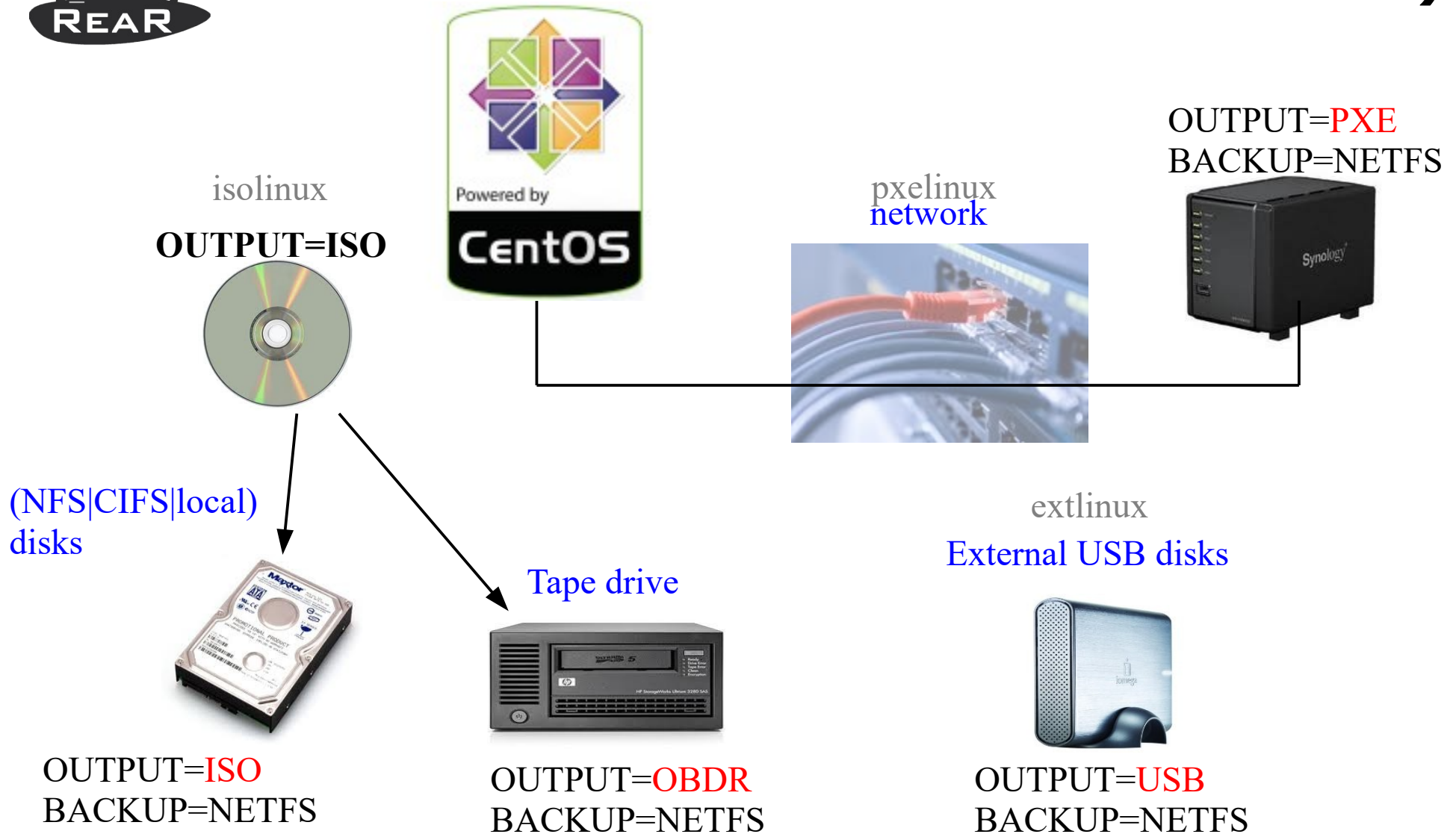


Relax and Recover - Output Flows

- **OUTPUT** defines the destination of the rescue image
- Valid OUTPUT types are:
 - ISO – creates a bootable ISO image
 - USB – creates a bootable USB disk/stick
 - PXE – creates a bootable PXE/TFTP image
 - OBDR – creates a bootable tape (ISO emulation)
 - RAMDISK – creates a bootable RAM image



BACKUP type *NETFS* (internal backup solution)





Location **BACKUP_URL**

- **BACKUP=NETFS**
- **BACKUP_URL** can be
 - File type: `BACKUP_URL=file:///directory/`
 - NFS type: `BACKUP_URL=nfs://nfs-server/directory/`
 - CIFS type: `BACKUP_URL=cifs://samba/directory/`
 - USB type: `BACKUP_URL=usb:///dev/sdc1/directory/`
 - Tape type: `BACKUP_URL=tape:///dev/nst0`
 - ISO type: `BACKUP_URL=iso:///directory/`

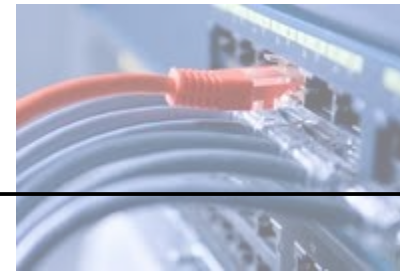


BACKUP type BAREOS (external backup solution)

Boot from
OUTPUT=**ISO**
BACKUP=BAREOS



Boot from
network



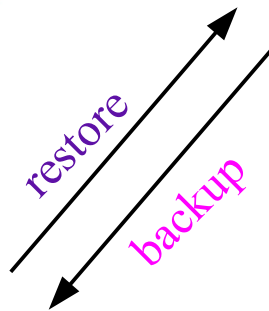
OUTPUT=**PXE**
BACKUP=BAREOS



Boot from
External USB disks



OUTPUT=**USB**
BACKUP=BAREOS



Bareos Backup
Server

Tape drive





`/etc/rear/local.conf`

- Define your settings in `/etc/rear/local.conf` (or `/etc/rear/site.conf`)
- `# grep -v -E '(^#|^$)' /etc/rear/local.conf`
`OUTPUT=ISO`
- Add:
`BACKUP=NETFS`
`BACKUP_URL=nfs://server/path`
- On NFS server backup => `/path/$(hostname)/`



Rear dump

- View system configuration:

```
# rear dump
```

```
Relax-and-Recover 1.17.2-git201512090939 / 2015-12-09
```

```
Using log file: /var/log/rear/rear-antwerp.log.lockless
```

```
Dumping out configuration and system information
```

```
This is a 'Linux-x86_64' system, compatible with 'Linux-i386'.
```

```
System definition:
```

```
ARCH = Linux-i386
```

```
OS = GNU/Linux
```

```
OS_MASTER_VENDOR = Fedora
```

```
OS_MASTER_VERSION = 7
```

```
OS_MASTER_VENDOR_ARCH = Fedora/i386
```

```
OS_MASTER_VENDOR_VERSION = Fedora/7
```

```
OS_MASTER_VENDOR_VERSION_ARCH = Fedora/7/i386
```

```
OS_VENDOR = RedHatEnterpriseServer
```

```
OS_VERSION = 7
```




Rear help

- Usage: `rear [-h|--help] [-V|--version] [-dsSv] [-D|--debugscripts SET] [-c DIR] [-r KERNEL] [--] COMMAND [ARGS...]`
- Available options:
 - **-h --help** **usage information**
 - **-c DIR** **alternative config directory; instead of /etc/rear**
 - **-d** **debug mode; log debug messages**
 - **-D** **debugscript mode; log every function call (via 'set -x')**
 - **--debugscripts SET** **same as -d -v -D but debugscript mode with 'set -SET'**
 - **-r KERNEL** **kernel version to use; current: '3.10.0-229.14.1.el7.x86_64'**
 - **-s** **simulation mode; show what scripts rear would include**
 - **-S** **step-by-step mode; acknowledge each script individually**
 - **-v** **verbose mode; show more output**



Rear help

- Usage: rear [-h|--help] [-V|--version] [-dsSv] [-D|--debugscripts SET] [-c DIR] [-r KERNEL] [--] COMMAND [ARGS...]
- List of commands:
 - checklayout check if the disk layout has changed
 - format format and label media for use with rear
 - mkbackup create rescue media and backup system
 - mkbackuponly backup system without creating rescue media
 - mkrescue create rescue media only
 - recover recover the system; only valid during rescue
 - savelayout save the disk layout of the system
 - shell start a bash within rear; development tool



Disaster Recovery in Practice

- Install rear software (*and required components*)
- Configure the `/etc/rear/site.conf` (or `/etc/rear/local.conf`)
 - `OUTPUT=`
 - `OUTPUT_URL=` (*optional; if not defined = `BACKUP_URL`*)
 - `BACKUP=`
 - `BACKUP_URL=` (*required for `BACKUP=NETFS`*)
- Make a system backup (*`rear mkbackup`*)
 - Create boot-able rescue media with system configuration (and optional with backup data)
- All steps are done “**online**”



Rear mkrescue

- Will create an ISO image stored as
 - `/var/log/rear/rear-$(hostname).iso`
 - On NFS server as `/path/$(hostname)/rear-$(hostname).iso`
- Inspect file `/var/lib/rear/layout/disklayout.conf`
- Try to boot from the ISO image into the RESCUE system
 - Use 'dmesg' to check if devices were found
 - Check the `/dev` directory to verify the storage devices were found



- [illegible]



Recovery Process in detail

- Boot system from rescue media
- Run: `rear -v recover`
- Creates `diskrestore.sh` script (*can be modified during run*)
 - Create partitions, RAID configuration and LVM
 - Create file systems (`mkfs`, `mkswap`)
 - Configure file systems (labels, mount points)
- Restore the backup data
- Restore the boot loader
- Inspect & Reboot



Recover with rear (1)

- Boot rescue image and select 'recover'

```
Relax-and-Recover v1.16.1-git201503121610

Recover linux-cgiq
Automatic Recover linux-cgiq

Other actions
Help for Relax-and-Recover
Boot First Local disk (hd0)
Boot Second Local disk (hd1)
Boot Next device
Hardware Detection Tool
ReBoot system
Power off system

Press [Tab] to edit, [F2] for help, [F1] for version info

Rescue image kernel 3.12.28-4-default Mon, 27 Apr 2015 15:18:44 +0200
BACKUP=NETFS OUTPUT=ISO BACKUP_URL=nfs://| | | | | /vol/ | | | | |
```



Get your hands dirty?

- We hope you want to dig deeper into rear!
- Getting started:
 - Use: `rear -s mkbakup`
to see the flow of the scripts it will execute
 - Depends on BACKUP method, architecture and OS version/brand
 - Be careful: `rear -s recover`
follows a different flow (seems logically, but you must understand the difference)



Where to put a script?

- **mkbackup** method: `/usr/share/rear/...`
 - `conf/` - configuration files (`/etc/rear/*.conf` read last)
 - `prep/` - preparation work; checking the environment
 - `layout/save/` - save the disk layout `/var/lib/rear/layout`
 - `rescue/` - modules, network, storage,...
 - `build/` - populate the initial ramdisk for our rescue image
 - `pack/` - create the initrd and copy kernel
 - `output/` - create the ISO image and copy to `OUTPUT_URL`
 - `backup/` - make the backup archive to `BACKUP_URL`



Where to put a script? (2)

- **recover** method: `/usr/share/rear/...`
 - `conf/` - read the configuration file + `/etc/rear/*.conf`
 - `setup/` - user defined scripts to run before recover
 - `verify/` - to check if a recover is possible at all
 - `layout/prepare` – recreate the disk layout
 - `restore/` - restore the archive from `BACKUP_URL`
 - `finalize/` - do some dirty tricks for disks, grub,...
 - `wrapup/` - copy the recover log to `/mnt/local/root/`



Relax and Recover (rear) Great Tool for your Disaster Recovery Team





Contacts

Web-site: <http://relax-and-recover.org/>

GitHub: <https://github.com/rear/rear>

Mailing list: rear-users@lists.relax-and-recover.org

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